When Do Mathematical Correspondences Show Up in Representational Similarity? A Controlled Study

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Abstract

Exact relationships between mathematical tasks offer a way to ask what a representational-similarity comparison identifies. We study single-task specialists on permutation statistics, directed-graph degrees, and binary matrix statistics. Two comparisons answer different questions: fixing the task pair while changing the input transformation, and fixing the transformation while changing the candidate task. In the primary PermWorld analysis, joint-answer-controlled final-layer CKA favors the correct over the specified wrong pairing in 26 of 32 conditions. The designated task exceeds the mean of 14 non-designated candidates in 29 of 32 conditions, but ranks uniquely first among 15 candidates in 19. Trained preference and change from initialization are distinct: only 18 conditions have a positive trained-minus-initial preference increment. Shared initial weights give the raw fixed-transformation task contrast a zero initialization reference, while candidate-specific answer stratification does not. A GRU task pair with near-perfect held-out accuracy illustrates how changing the encoding can reverse the increment without reversing trained preference. Graph and two independently sampled matrix worlds reveal further control and metric dependence; a matrix transpose pair with perfect measured task accuracy has negative size-averaged, answer-controlled CKA preference but positive neighborhood and Procrustes contrasts. These exploratory findings support correspondence-associated geometric selectivity under specified conditions, rather than a universal or uniquely mathematical mechanism. They also motivate reporting initialization, both pairing contrasts, answer controls, and prescribed measurement sensitivities together.

1 Introduction

Representational similarity is often used to argue that neural networks have learned a common underlying structure. A large similarity value, however, can reflect several different sources: the inputs used to measure it, features already present at initialization, properties of the output task, or a training-dependent organization of hidden states. This ambiguity is especially relevant when the proposed shared structure is a mathematical relationship between tasks. If two answers become equivalent after an exact input transformation, does the same transformation also produce a distinctive correspondence between their learned representations?

We investigate this question using tasks with explicitly checkable formulas. For example, the number of descents of a permutation equals the number of inverse descents of its inverse. Such identities identify a designated task pair and a correct transformation without selecting them from a similarity plot. They do not, by themselves, predict that two learned hidden spaces must become identical, or that every similarity measure must favor the designated relationship. The tasks may have different difficulty, the input encoding may make one transformation easier to express than another, and the same output identity may appear in several kinds of hidden geometry.

Our main study, PermWorld, uses sixteen permutation-statistic specialists, organized into eight directed correspondences. We compare one-line and canonical-cycle encodings within two existing model configurations, a Transformer and a GRU. Each configuration uses saved, genuinely shared initial weights and a fixed training endpoint. We also examine a small directed-graph study and two independently sampled binary-matrix worlds. These additional settings delimit the observation; their different task families and sampling distributions do not form a controlled domain intervention.

The central distinction is between two comparisons. First, a correct transformation can produce more similar representations than a specified wrong transformation for the same two models. Second, under a fixed transformation, the designated task model can be more similar than models trained for other tasks. Neither comparison implies the other. Nor does a positive preference after training imply a positive change from initialization. We keep these quantities separate throughout the study.

The paper makes three contributions:

1. A paired measurement design separating input-pairing selectivity from task-candidate selectivity, with an explicit shared-initialization reference and a decomposition of training increments into correct and wrong scores.

2. A complete exploratory PermWorld grid showing both the extent and the limits of correspondence-associated selectivity across encodings, configurations, layers, and controls. We report both above-mean and first-rank candidate counts, retaining all negative conditions.

3. Cross-system examples that show why answer controls and alternative readouts matter even when source tasks are well learned, together with concrete guidance for similarity-based arguments about shared structure.

Our claim concerns measured geometric preferences. We do not use these measurements to identify an internal mathematical algorithm or establish a functional transfer benefit.

2 Related work

Task similarity and task taxonomies. Relationships between training objectives provide an established explanation for representational similarity. Li et al. (2025) manipulate dataset and task overlap, finding greater similarity with greater overlap across several model settings. Their constant-data experiment changes the labeling of identical images to separate task overlap from shared examples. Dwivedi & Roig (2019) use representational similarity analysis between visual task models to construct a task taxonomy and examine associations with transfer performance. These studies motivate examining how task definitions shape geometry. The present study examines single-task specialists with exactly specified mathematical correspondences. One comparison fixes both specialists and changes the input pairing among correct, originally specified wrong, and identity transformations. A second fixes the left specialist, transformation, and anchors while varying the right task. Together, these comparisons ask where a designated correspondence has a geometric preference relative to explicit input-pairing and task-candidate comparators.

Representational convergence. Huh et al. (2024) propose the Platonic Representation Hypothesis and report increasing agreement between vision and language representations with model scale. Park (2026) study geometric tasks over a shared world of city coordinates. Multi-task training aligns representations even for non-overlapping task sets, while some tasks impair the integration of new entities through fine-tuning. This distinction between aligned geometry and adaptability is relevant here. The current experiments preserve individual correct and wrong scores, trained pairing preferences, and initialization preferences, so that an increment can be distinguished from an increase in correct-pair similarity. Their scope is exploratory correspondence-associated geometry; functional benefit requires separate evidence.

Grounding and calibrating similarity measurements. CKA supplies invariances useful for comparing representations (Kornblith et al., 2019), but a similarity value requires an appropriate reference. The

ReSi benchmark evaluates measures using prediction-grounded and design-grounded tests, reflecting multiple meanings of model similarity (Klabunde et al., 2025). Cui et al. (2022) show that input population structure can inflate RSA and CKA, including comparisons of random networks, and propose covariate-adjustment regression. Gröger et al. (2026) identify model-scale confounding and introduce permutation-null calibration, separating global spectral convergence from local neighborhood agreement. These results motivate retaining input, initialization, answer, and output references and checking prescribed alternative metrics. The present trained-minus-initial preference contrast has its own definition and is distinct from permutation calibration. Answer stratification also retains pair-specific supports and cannot remove every confidence or nonlinear answer contribution. No subtraction of hidden and output scores is interpreted as an isolated internal mechanism.

Permutation encodings and preceding work. Scullen et al. (2025) use permutation statistics to study how encoding, data availability, and MLP or Transformer models affect learning. Their five encodings comprise permutation matrices, inversion vectors, Lehmer cocodes, major codes, and lexicographically minimal reduced words. They identify internal-representation comparison as a direction for further study. The present one-line and canonical-cycle encodings support an additional comparison of initialization and trained pairing preferences, with all declared layers and relations retained. Model capacity and task proficiency differ across configurations, limiting a pure architecture interpretation. The GraphWorld and MatrixGroupWorld analyses examine the scope of the same comparisons in additional mathematical settings, with their negative, undefined, and failed verification results preserved.

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3 Study design and two distinct comparisons

3.1 What each comparison answers

Let f be a left task, g_* its designated right task, and T an input transformation. A relation has the form

$$f(x) = \phi\big(g_*(T(x))\big), \tag{1}$$

where ϕ is a specified answer-coordinate map. It is the identity for most PermWorld relations, a shift for one permutation relation, and a complement map for graph-degree complement relations. These formulas are verified against the task generators. They are specified independently of the geometric scores.

Comparison A: fix the two tasks, change the input pairing. We compare $h_f(x)$ with $h_{g_*}(T(x))$ and with $h_{g_*}(U(x))$, where U is the original specified wrong transformation. The question is whether the designated transformation is preferred over that comparator for these same two trained specialists. The identity pairing is also retained. A wrong transformation need not break the answer identity on every input.

Comparison B: fix the input transformation, change the right task. We instead keep the left model, x, and $T(x)$ fixed and compare $h_f(x)$ with $h_g(T(x))$ for every available $g \neq f$. The question is whether g_* stands out among candidate task models under the same input transformation. In PermWorld this gives fifteen candidates, including the designated task, and fourteen non-designated comparators. GraphWorld and MatrixGroupWorld each give three candidates. A non-designated task is not asserted to lack all mathematical relationships with f. Task identity, output statistics, and proficiency also differ between these candidates.

3.2 Scores, preferences, and training increments

For Comparison A, with $s \in \{0, t\}$ denoting saved initialization and the fixed trained endpoint, define

$$\begin{aligned} S_c^s &= S\big(h_f^s(x), h_{g_*}^s(T(x))\big), & S_u^s &= S\big(h_f^s(x), h_{g_*}^s(U(x))\big), \\ B_0 &= S_c^0 - S_u^0, & B_t &= S_c^t - S_u^t, \\ \delta_c &= S_c^t - S_c^0, & \delta_u &= S_u^t - S_u^0, \end{aligned} \tag{2}$$

$$\Delta = B_t - B_0 = \delta_c - \delta_u. \tag{3}$$

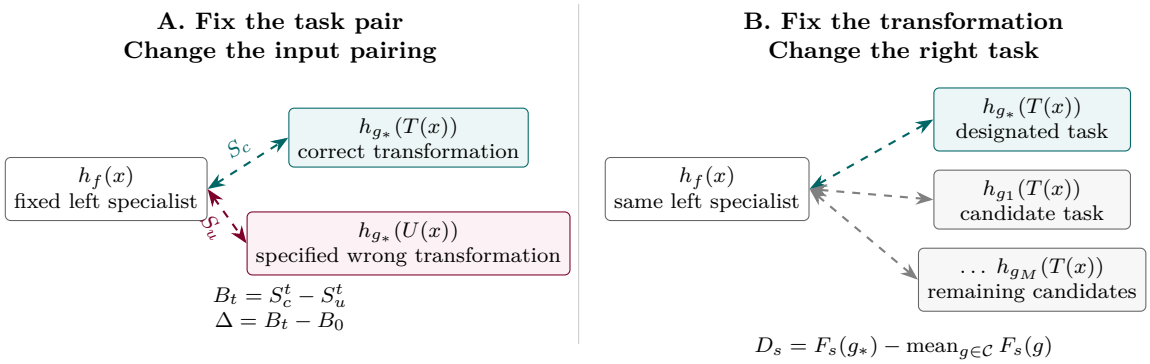


Figure 1: Two comparisons with different fixed factors. Left: the same pair of specialists is evaluated under the correct T and specified wrong U; B_t contrasts these two similarities. Right: the input transformation stays T while the right task varies; D_t contrasts the designated specialist with a prespecified candidate category. Dashed links denote similarity comparisons, not learned computational mappings. All states are taken before the task token. Shared initial weights give the *raw* right-hand contrast a zero reference; candidate-specific answer controls can change that reference.

We preserve the four original scores as well as these contrasts. Positive Δ can come from a reduction of wrong-pair similarity without an increase of correct-pair similarity. Changing an initialization reference changes Δ, not an already fixed B_t.

For Comparison B, use separate notation,

$$F_s(g) = S\big(h_f^s(x), h_g^s(T(x))\big), \qquad D_s = F_s(g_*) - \frac{1}{|\mathcal{C}_s^{\mathrm{fin}}|}\sum_{g\in\mathcal{C}_s^{\mathrm{fin}}} F_s(g), \tag{4}$$

where $\mathcal{C}$ is the prespecified comparator category and $\mathcal{C}_s^{\mathrm{fin}}$ its finite, complete-length candidates at phase s. The contrast is undefined if the designated score is undefined or this set is empty. We report every candidate and all finite/undefined counts. In the main PermWorld grid, all fifteen candidates are finite in both phases. An optional training change recomputes both means on the common set $\mathcal{C}_0^{\mathrm{fin}} \cap \mathcal{C}_t^{\mathrm{fin}}$; it is a different contrast from Δ in Eq. 3.

The primary score is ordinary linear CKA (Kornblith et al., 2019). For centered sample-by-feature matrices X and Y,

$$\mathrm{CKA}(X,Y) = \frac{\|X^\top Y\|_F^2}{\|X^\top X\|_F\,\|Y^\top Y\|_F}. \tag{5}$$

Zero-variance or otherwise undefined measurements remain undefined. The other final-layer readouts are ten-neighbor overlap and negative held-out normalized Procrustes error; their larger-is-more-similar convention is fixed, but their scales and normalizations differ. They are sensitivities on the same representations, not independent replications or functional measurements.

3.3 Shared initialization and its exact scope

Within a PermWorld architecture, all sixteen specialists and both encodings start from *the same numerical initial parameters*, copied from a saved step-0 state. This is stronger than reusing a seed. Checkpoint and parameter hashes bind the initial identities, including the recorded value-preserving GRU key conversion. GraphWorld copies the same saved Transformer initialization used in PermWorld. In MatrixGroupWorld the four specialists within each of two units share that unit's saved initial weights; the units themselves use independently sampled data and initials.

The measurement point is the end of the encoded input, before the task token, with models in evaluation mode and dropout disabled. Therefore, under shared weights, a right specialist's *raw* initial state at $T(x)$

System	Tasks / trained sources	Configurations and input sizes	Units and initial-weight sharing	Comparison coverage
Perm	16 / 64	Transformer, GRU; two encodings; $n = 10, \ldots, 30$	One paired data world; one actual initial per architecture, copied across tasks and encodings	A: $8 \times 4 = 32$; B: $32 \times 15 = 480$ candidates
Graph	4 / 4	Transformer; off-diagonal adjacency tokens; $n = 6, 7, 8$	One world; PermWorld Transformer initial reused	A: 4; B: $4 \times 3 = 12$ candidates
Matrix	4 / 8	Transformer; row-major binary tokens; $n = 6, 8, 10$	Two independent data/initial units; four tasks share weights within each unit	A: $4 \times 2 = 8$; B: $8 \times 3 = 24$ candidates

Table 1: Coverage, rather than independent replication counts. All 76 source models have actual fixed 20k endpoints. Comparison B uses ordinary final-layer CKA only. In PermWorld, layer analyses use CKA and its prescribed debiased sensitivity; the three principal geometric readouts are jointly available at the final state. Shared tasks, transformations, models, inputs, and initials create dependence within these grids.

is independent of its eventual task g. With a common anchor set and task-independent preprocessing, and when S is defined and the comparator category is nonempty,

$$h_g^0(T(x)) = h_0(T(x)) \quad \Longrightarrow \quad F_0(g) = F_0(g') \quad \Longrightarrow \quad D_0 = 0. \tag{6}$$

Consequently the raw fixed-transform task preference in this design arises after training. This statement does not isolate the mathematical relation from other effects of task training. It also does not apply to task-conditioned outputs, or to answer-controlled scores with candidate-specific residualization.

For Comparison A the correct and wrong inputs already differ at step 0; B_0 is generally nonzero even though the models share weights. These different initialization references explain why the two comparisons give different answers to "did the preference emerge through training?"

3.4 Systems, training, and coverage

Table 1 separates model coverage from independent sampled units. PermWorld supplies sixteen statistics and eight exact directed relations (Appendix A). The two input encodings describe the same permutations: one-line notation and cycle notation normalized by cycle rotation and ordering. All elements, including fixed points, are represented. The two configurations share vocabulary, width 256, four layers, and dropout 0.1. The Transformer has eight heads and approximately 3.24 million parameters; the GRU has approximately 1.66 million. Their capacities are not matched.

Each task is trained separately to emit an answer token and EOS. No representation-alignment loss or relation-supervised operator is used. The common declared budget is 20,000 updates with effective batch size 64, giving 1,280,000 example exposures per specialist. Endpoints are fixed, not selected for geometric results or best validation accuracy. Within a world, data and batch schedules are shared across tasks; encoding pairs also share their recorded starting random state. Sixteen existing PermWorld endpoints were reused and forty-eight were completed later with documented source and engineering-repair histories. These histories are preserved rather than claiming that all executions used one byte-identical program revision. Appendix C gives optimization and random-stream details.

All systems use 4,096 training, 256 validation, and 512 held-out anchors per declared size. Splits follow their documented transform-orbit exclusions. In PermWorld only accepted eight-element orbits are used; matrices

require exact full rank over $\mathbb{F}_2$ and exclusion of overlapping embedded $\mathrm{GL}(2, \mathbb{F}_2)$ action/transpose orbits. These conditional sampling rules do not imply uniform sampling over all mathematical objects.

3.5 Controls and aggregation

CKA is calculated separately within every size, then aggregated with fixed weights 1/21 for PermWorld and 1/3 for GraphWorld and MatrixGroupWorld. Subtracting length means and pooling all lengths is not substituted for this procedure. Any undefined size component leaves its aggregate undefined; weights are not redistributed.

The primary control subtracts means within joint gold-answer strata. For a candidate g, the key records size, the left identity answer, and mapped right answers on the identity, correct, and wrong views. A stratum requires at least three samples separately in the fitting and held-out splits. The same retained anchors and groups are used across phases and input roles for a given pair. In Comparison B, changing g changes these labels and can change both groups and support. Writing its residualization operator as R_g makes the scope of Eq. 6 explicit:

$$F_0^{\text{answer}}(g) = \mathrm{CKA}\big(R_g H_0(x), R_g H_0(T(x))\big) \quad \text{need not be constant in } g. \tag{7}$$

This is a control for between-group answer geometry, not removal of all answer information, confidence, or prediction-error structure. We retain true-answer and output-probability baselines, and do not interpret hidden minus output contrasts as a mechanism estimate. GraphWorld's additional answer-plus-exact-edge-count control remains a separate sensitivity.

Procrustes maps are fitted only on validation features, with a prescribed PCA rank cap of eight, RMS scaling, and orthogonal alignment; held-out features are centered according to the original protocol. Neighbor overlap uses $k = 10$ with stable anchor-ID tie rules and the original duplicate exclusions. We retain numerical failures and undefined values. The fixed candidate catalogue distinguishes designated, other-known-exact, and non-designated correspondences before their scores were evaluated; the other-known-exact category is empty in the available PermWorld candidate catalogue, not a proof of absence of other relations.

4 Results: three questions

All results below are exploratory. The candidate analysis was frozen before its scores were computed, after earlier training and geometric results had been seen. We therefore do not describe it as an independently preregistered confirmation. Counts summarize coupled conditions; they are not sample sizes for a binomial test.

4.1 Question 1: do designated correspondences have geometric selectivity?

In Comparison A, joint-answer-controlled final-layer ordinary CKA yields $B_t > 0$ in 26 of the 32 PermWorld conditions (Figure 2). The other six conditions are all canonical-cycle encodings: descents/inverse descents, fixed/anti-fixed points, and global descents/components for the Transformer; fixed/anti-fixed points, increasing/decreasing runs, and global descents/components for the GRU. This concentration is a descriptive encoding association, not evidence that encoding is the sole cause or that the negative cases are noise.

Comparison B asks a different task-identification question while holding the transformed input fixed. Under the same primary answer-controlled readout, the designated specialist exceeds the mean of the fourteen non-designated specialists in 29/32 conditions, but is uniquely first among all fifteen candidates in 19/32 (Figure 3). All fifteen candidates are finite in these main conditions. Above-mean preference is therefore more frequent than uniquely identifying the designated model. Table 2 also gives the raw readout, so that this observation is not tied to reporting only the favorable control.

For context, if the designated identity were exchangeable among fifteen untied scores, its chance of ranking first would be $1/15 \simeq 6.7\%$. The observed 19/32 (59.4%) is well above this *descriptive reference*. Task labels, distributions, and learning difficulties are not exchangeable here, and conditions share models and inputs. We therefore do not attach a binomial p-value or treat 1/15 as a validated experimental null.

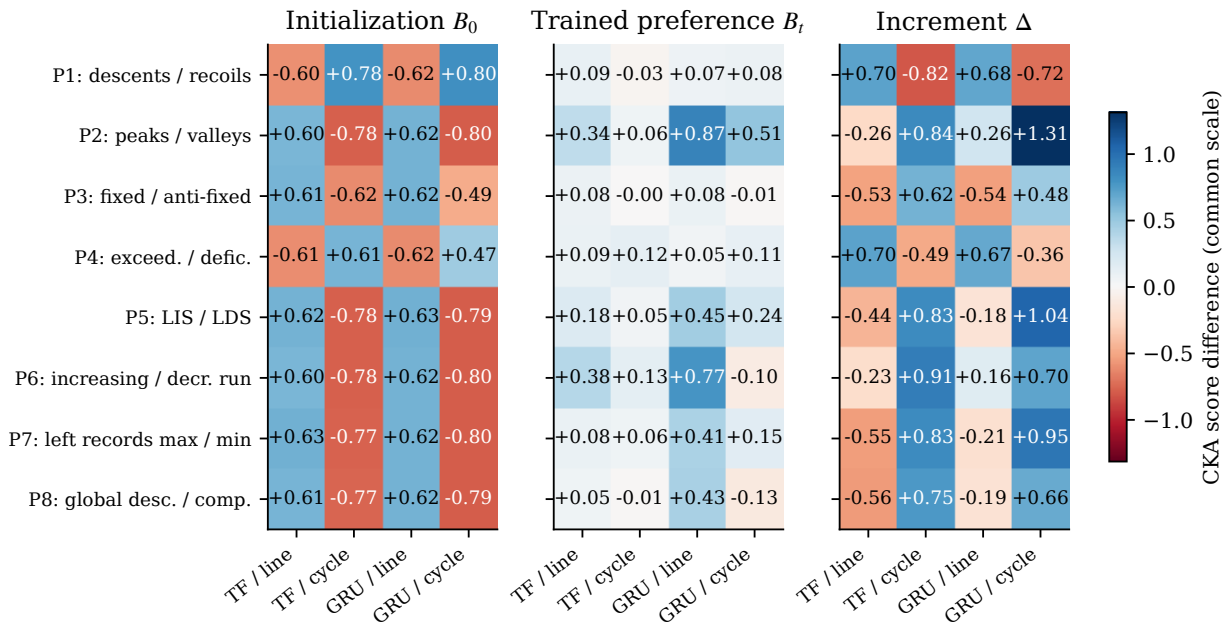


Figure 2: All eight PermWorld relations and four configuration/encoding conditions under joint-answer control, final normalization, and equal-within-length ordinary CKA. Initialization preference B_0, trained preference B_t, and increment Δ use a common color scale. Positive trained preference occurs in 26/32 conditions, but positive increment in 18/32. The raw scores and their correct/wrong changes are retained in Appendix B; task IDs P1–P8 are defined in Table 5. This figure does not summarize other layers or controls.

	A: vary input pairing		B: vary right task	
Control	$B_t > 0$	$\Delta > 0$	Above other mean	Unique first of 15
Raw, within length	27/32	25/32	30/32	25/32
Joint-answer strata	26/32	18/32	29/32	19/32

Table 2: Different questions and different readouts. The mean in Comparison B contains fourteen non-designated tasks; the ranking includes the designated task. All rows use final-layer ordinary CKA and the original equal 21-length weights. Counts are descriptive, dependent cells.

4.2 Question 2: what changed through training, and what changed with encoding?

The shared-initialization reference answers this question differently for the two comparisons. In the raw version of Comparison B, candidate initial scores are identical within every fixed condition; stored designated-minus- mean contrasts differ from zero by at most 1.4×10^{-17} from mean summation. Thus raw designated-task advantages arise after the specialists have been trained. In the answer-controlled version, none of the 32 designated-minus-mean initial contrasts is exactly zero: four are positive and twenty-eight negative, ranging from -0.2477 to $+0.0935$. This difference is consistent with candidate-specific groups and support, not a failure of initial-weight identity.

In Comparison A, B_0 is already influenced by the geometry of T and U on each encoding. Accordingly, trained preference is positive in 26/32 conditions while Δ is positive in only 18/32. A negative increment does not imply a negative final preference. Conversely, a large positive increment may be driven primarily by the wrong score decreasing. For the one-line Transformer descents/inverse-descents pair, correct answer-controlled CKA changes from 0.0660 to 0.1191, while wrong CKA changes from 0.6709 to 0.0260. The resulting $\Delta = 0.6980$ contains a 0.6449 contribution from the wrong-score decline and only a 0.0531 correct-score increase. It measures a reshaping of selectivity, not a 0.6980 increase of correct-pair agreement.

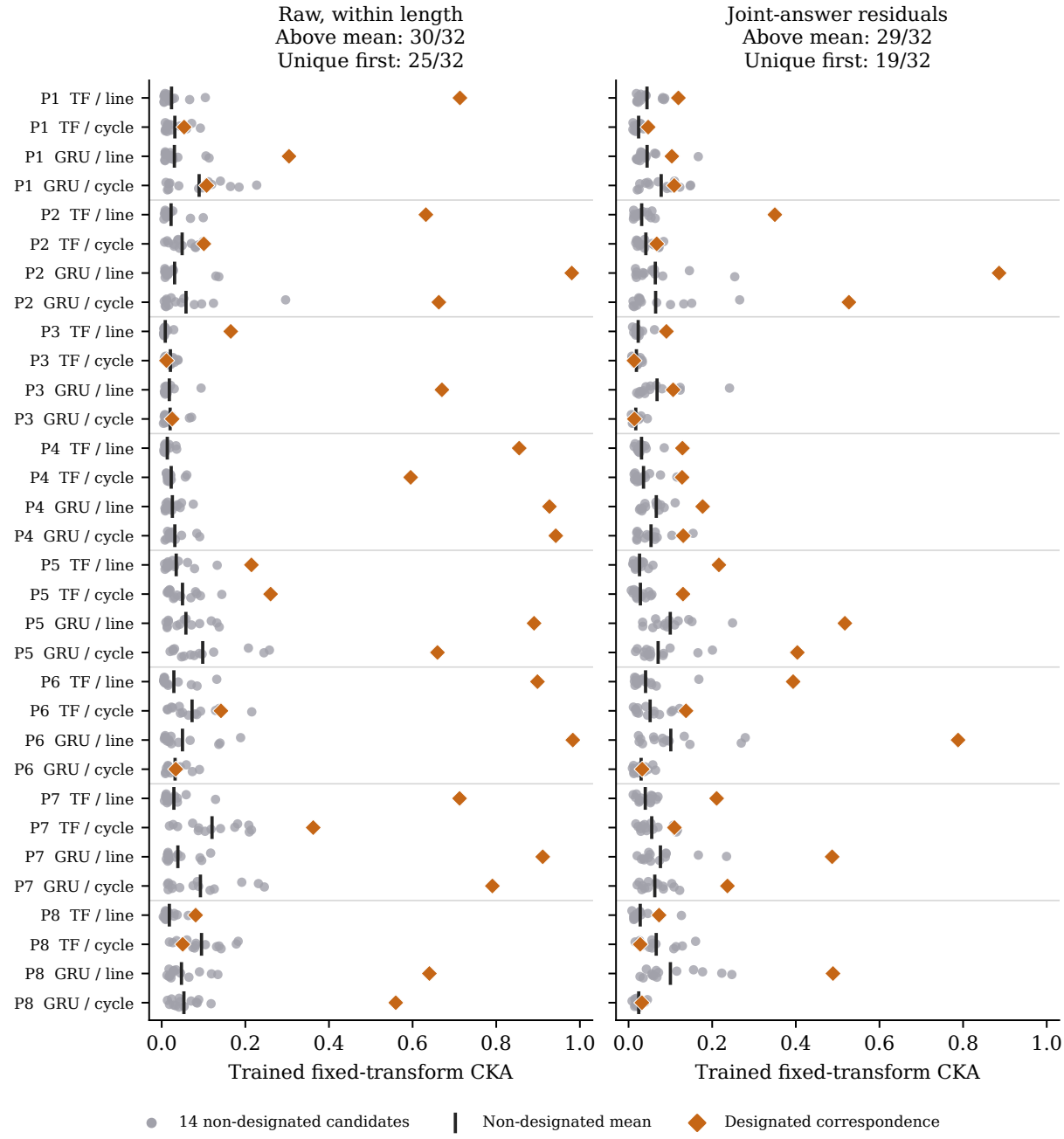


Figure 3: Every trained fixed-transform candidate score for the 32 PermWorld conditions. Each row fixes the left specialist, transformation, input anchors, encoding, and configuration. Gray points are the fourteen non-designated specialists, black ticks their mean, and orange diamonds the designated specialist. Raw and answer-controlled scores are separate. The latter use candidate-specific strata despite sharing the underlying anchor pool. Initial raw candidate scores are identical, but initial answer-controlled candidate scores need not be. Neither these 480 logical candidate pairings nor the two controls are independent replications.

GRU encoding	B_0	B_t	Δ	Left acc.	Right acc.
One-line	-0.6230	$+0.0455$	$+0.6685$	99.88%	99.30%
Canonical cycle	$+0.4710$	$+0.1098$	-0.3612	100.00%	100.00%

Table 3: Exceedances/deficiencies, joint-answer-controlled final CKA. Accuracy is the original equal-length held-out identity-input answer accuracy over the full vocabulary. Task proficiency is high in both encodings, yet the initial preference and training increment differ. The trained preference has the same sign.

The GRU exceedances/deficiencies pair supplies a case where weak learning is not an adequate account of an encoding-dependent increment (Table 3). Both tasks attain approximately 99–100% held-out answer accuracy in both encodings. The trained preference stays positive, increasing from 0.0455 to 0.1098, whereas Δ changes from $+0.6685$ to -0.3612 because B_0 changes from -0.6230 to $+0.4710$. This is an increment reversal without a reversal of trained relationship preference.

More generally, for an encoding comparison define $\Gamma_0 = B_0^{\text{cycle}} - B_0^{\text{line}}$ and $\Gamma_t = B_t^{\text{cycle}} - B_t^{\text{line}}$. Then $\Gamma_\Delta = \Gamma_t - \Gamma_0$. Reporting only Γ_Δ would conflate a change of initialization preference with a change of trained preference. All thirty-two individual correct and wrong changes remain available in Appendix B.

4.3 Question 3: which measurement and system dependencies remain?

Layers and readouts in PermWorld. At the primary final state, ordinary CKA, neighbor overlap, and held-out Procrustes have matching B_t directions in 32/32 conditions and matching Δ directions in 30/32. The two increment disagreements are the one-line GRU peaks/valleys and increasing/decreasing-run pairs: CKA and Procrustes have positive Δ, but neighbor overlap has negative Δ. Their exact values are retained in the released tables. These are geometrically different readings of the same activations, not three independent reproductions. Multilayer CKA is also not uniformly selective (Figure 4). For example, the GRU has positive answer-controlled B_t in 8/16 conditions at layer 1 and 13/16 at layer 4 and final normalization. The Transformer changes from 10/16 to 13/16. The input-end embedding score is undefined in all sixteen conditions of each architecture under this control; it cannot serve as a numerical zero or a useful measurement of pooled input content. Changes between layer 4 and final normalization reflect normalization, including learned affine parameters, rather than a fifth Transformer or recurrent block.

Output references. The numeric-output probability baseline also contains pairing preferences: raw ordinary CKA has positive B_t in 29/32 conditions and positive Δ in 30/32; after answer control these counts are 29/32 and 16/32. It is evaluated at a task-conditioned output point, unlike the primary pre-query hidden state. Gold-answer stratification does not remove all confidence and prediction-error geometry. These parallel readouts therefore remain relevant alternative explanations, not quantities that can be subtracted from hidden scores to estimate an additional mechanism.

Directed graphs and answer structure. GraphWorld uses maximum/minimum in- and out-degree specialists and exact edge-reversal or complement correspondences. Raw final CKA gives positive B_t and Δ in all four relations; after joint-answer control, B_t is positive in two and Δ in none. The contrast therefore depends substantially on the measured answer-related geometry. In Comparison B the raw designated task is uniquely first in all four relations; under answer control it exceeds the other-task mean in three and is uniquely first among three candidates in two. These are again different contrasts, with the full twelve-candidate table retained. Answer-plus-edge-count stratification is retained as a separate sensitivity; its changed groups do not automatically make it the uniquely correct readout. The four original wrong-transform comparators happen to satisfy their designated answer formulas on 54.8–59.7% of raw analysis anchors. These actual-sample rates differ from the earlier random mathematical audit. We retain those rows and comparators, limiting the claim to the specified contrast rather than a perfect relation-breaking intervention. GraphWorld also shares the PermWorld Transformer initial, so it is not independent initialization replication.

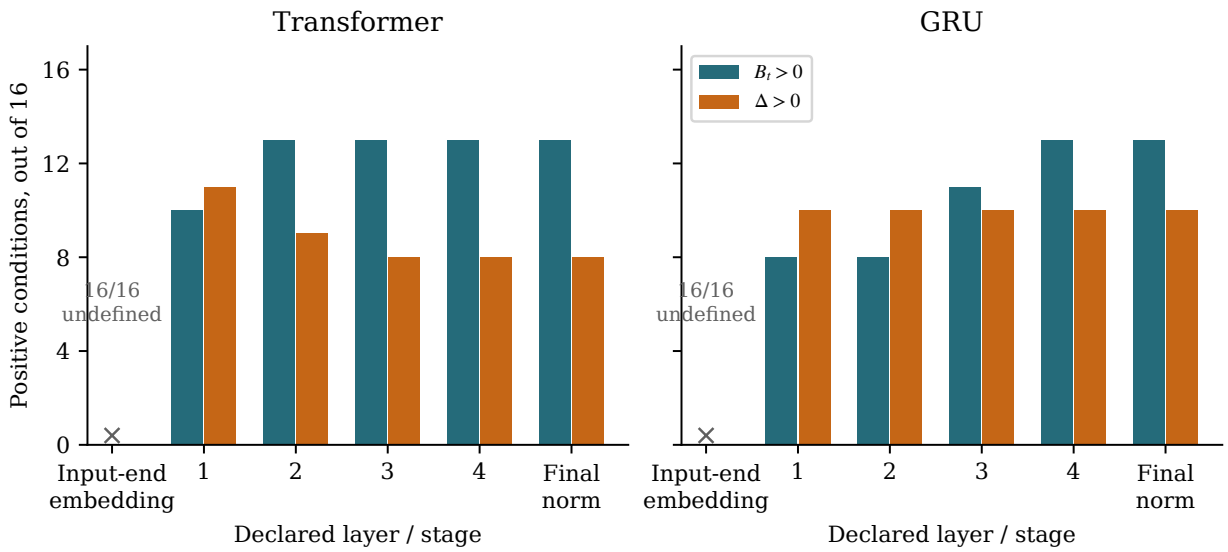


Figure 4: Prescribed layer/stage CKA readouts on all eight relations and both encodings within each architecture: sixteen coupled conditions per panel. Bars count positive B_t and Δ under the same answer control and fixed length weights. Input-end embedding scores are all undefined, not zero. Neither the most favorable layer nor a different subset is selected as a new primary analysis. Final-layer kNN/Procrustes agreement is a separate sensitivity and is not implied for every plotted layer.

4.3.1 Independent paired worlds

MatrixGroupWorld provides two independently resampled worlds, each with new data and a new initialization. Within each world, four task specialists share a saved, genuine initialization and each completes the fixed 20,000-update budget. We report both worlds separately, with equal weights over $n \in \{6, 8, 10\}$ and the same prespecified transformations, controls, and metrics. These worlds repeat the matrix setting; they do not independently confirm PermWorld across initializations. Because data and initialization vary together, differences between worlds do not isolate a causal effect of initialization. The four relations, three sizes, and three metrics are coupled measurements rather than additional independent replications.

4.3.2 R1: an illustrative case

R1 relates F_1 and F_2 through transposition. Both tasks achieve 100% full-vocabulary answer accuracy in both worlds at every size on the complete identity training, validation, and analysis sets. The F_2 models also achieve 100% analysis accuracy under the correct and wrong R1 transformations. Nevertheless, their geometric pairing preference depends on the control and metric (Figure 5; Table 4). This makes R1 an illustrative case, rather than evidence for a general reversal of relation preference.

For raw, size-stratified hidden-state CKA, worlds one and two respectively give $(B_t, \Delta) = (0.520209, 0.519581)$ and $(0.558791, 0.553006)$. After joint-answer-stratum control, these become $(-0.019384, -0.020747)$ and $(-0.014022, -0.014551)$. Thus, these controlled, equally weighted CKA estimates favor the wrong pairing. Answer control changes both centering and retained sample support; this contrast cannot be interpreted as a causal subtraction of all answer information.

The fixed-transform comparison supplies an additional distinction. For R1, the designated task ranks uniquely first among three candidates under answer control in both worlds, both initially and after training. Its advantage over the two non-designated tasks changes from 0.1894 to 0.1182 and from 0.1179 to 0.0663. Thus first rank in Comparison B can coexist with a negative input-pairing preference in Comparison A, and its controlled task advantage need not originate at training.

	Matrix unit 1		Matrix unit 2	
Answer-controlled R1 readout	B_t	Δ	B_t	Δ
CKA	-0.019	-0.021	-0.014	-0.015
10-neighbor overlap	$+0.048$	$+0.040$	$+0.082$	$+0.070$
Negative Proc. error	$+0.052$	$+0.039$	$+0.045$	$+0.036$

Table 4: R1 as an illustrative disagreement case. Both column/row-weight specialists achieve 100% identity-input accuracy at each size in training, validation, and analysis. Equal-n answer-controlled CKA preferences are negative in both units, while neighbor and negative-error Procrustes preferences are positive. Each readout has its own scale. This table does not assert agreement at every individual n.

The controlled sensitivity metrics give different aggregate directions. For kNN overlap, (B_t, Δ) is $(0.047718, 0.039648)$ and $(0.081837, 0.069883)$; for held-out Procrustes it is $(0.052454, 0.038940)$ and $(0.045210, 0.036150)$. Procrustes is scored as negative alignment error, so larger values indicate better alignment. These metrics use different scales; their numerical magnitudes are not directly comparable. Raw B_t and Δ are positive for all three metrics in both worlds, whereas controlled CKA disagrees with the two sensitivity metrics. This aggregate disagreement also has size-specific qualifications: controlled CKA has positive B_t and Δ at $n = 10$ in both worlds, while controlled Procrustes has negative B_t and Δ at $n = 6$ in both. World one's controlled kNN B_t and Δ are negative at $n = 8$. We retain all sizes rather than select a uniformly positive or negative subset.

R3 illustrates why Δ must also be decomposed. Its controlled CKA increments are 0.048498 and 0.121332, with correct-pair changes $\delta_c = -0.800773$ and -0.755736, and wrong-pair changes $\delta_u = -0.849270$ and -0.877068. Both scores decrease; the larger wrong-pair decrease produces the positive increment. These results describe a change in relative selectivity, not increased correct-pair convergence. Interpretation is further limited by weak F_3 learning at $n = 10$: training accuracies are 45.61% and 39.21%, validation accuracies 40.23% and 31.25%, and identity-analysis accuracies 35.55% and 31.64%. Accuracy on the correct R3-transformed analysis inputs remains 35.74% and 32.23%. The low training accuracy prevents attributing this weakness solely to generalization.

We interpret these observations as measured pairing selectivity, with algebraic and causal mechanisms unresolved. Numerical checking retains its original verification caveat.[1]

5 Discussion and implications for similarity-based arguments

The observations support a conditional relationship between exact task correspondences and geometric pairing preferences. In PermWorld the designated specialist often exceeds a task-candidate mean and frequently ranks first. The same complete grid also contains negative preferences, increment reversals, and layer-dependent readings. The graph and matrix studies further show that answer control and metric choice can change the answer to an otherwise precisely defined comparison.

Three interpretive distinctions are useful. First, a trained preference describes a trained geometry, whereas an increment describes that geometry relative to a specified initial reference. Second, preferring a transformation for a fixed task pair does not uniquely identify the task pair under a fixed transformation. Third, an answer-controlled geometric association is not an identified mathematical mechanism. Pair-specific residualization, model proficiency, output uncertainty, and general state organization remain plausible contributors. R1 demonstrates a metric and control discrepancy after successful learning; it does not settle why that discrepancy occurs.

For practitioners using similarity to argue that models share a structure, our design suggests four reporting practices:

[1] The unchanged audit `c2e799...` records 64 passed, two failed, and six consistently undefined checks at tolerance 10^{-8}. Both failures concern trained raw output-probability kNN, for correct and wrong pairings, on the fixed 64-input first-world R1/$n = 6$ subset. They remain failed; see Appendix D.

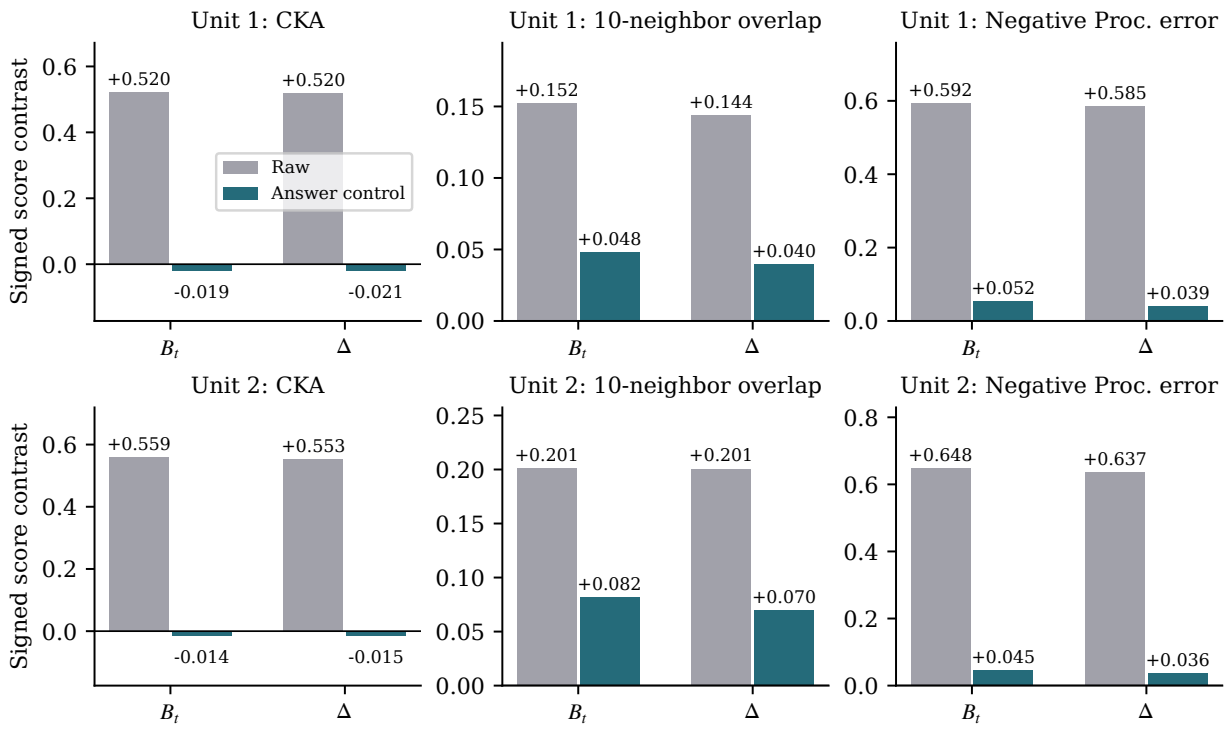


Figure 5: Matrix R1 raw and answer-controlled preferences and increments, in each of two independent data/initialization units. Control changes the direction of the aggregated CKA contrast, while the other two controlled readouts remain positive. Axes are metric-specific; differences of magnitudes across metrics do not share a common normalization. The example is one relation in two units and is not a general cross-system law. Original numerical verification failures on a separate output-probability kNN subset remain reported in Appendix D.

1. Show trained selectivity and its actual initialization reference, including correct and comparator score changes separately.
2. Under the same transformation, retain a full prespecified candidate task set and distinguish above-average agreement from uniquely identifying the designated correspondence.
3. Report raw, answer-controlled, and output-reference geometry, together with group/support changes and undefined measurements.
4. Keep the designated layer primary while showing prescribed layer and metric sensitivities; disagreements constrain the claim rather than providing a reason to select a favorable readout.

These practices apply beyond mathematical testbeds whenever representation similarity is used as evidence about shared task structure. They are a measurement discipline; the present experiments do not establish that every application will show the same positive preferences.

6 Limitations

Shared initials and limited independent replication. The principal PermWorld result uses one data world and one actual initialization per architecture, copied across specialists and encodings. GraphWorld reuses the Transformer initial. Shared weights make Eq. 6 interpretable, but do not show that the same preferences hold when specialist initializations are independent, or across new PermWorld data/initial units. This is the foremost replication gap. The two matrix worlds independently vary data and initialization

together; they do not isolate the effect of initialization and do not confirm the PermWorld 32-condition count. Independent paired worlds and independently initialized specialists would answer different follow-up questions.

Configuration, proficiency, and encoding. Transformer and GRU capacities differ, so their comparison is between the declared configurations rather than a pure architecture effect. Several cycle-encoding or matrix-XOR tasks remain weak. We retain their scores and training-defined modal baselines instead of requiring favorable geometry or a post hoc accuracy threshold. Exceeding a length mode does not imply full task mastery, and low aggregate accuracy does not prove that a model ignores all input content. The high-proficiency GRU example and R1 constrain a universal learning-failure account but do not remove proficiency confounding elsewhere. Matching updates and example exposures also does not match FLOPs, wall time, or the number of input tokens processed. Encoding changes tokenization and positions; the present intervention does not separate every component of encoding.

Control support, dependency, and sampling. Answer residualization varies by candidate and does not remove all nonlinear label or confidence information. Wrong transformations have accidental answer matches. Tasks share inputs, weights, schedules, and transformation families, and the eight permutation relations are not eight independent scientific replications. We report deterministic summaries on the fixed analysis samples without inference treating pairs, lengths, metrics, or conditions as independent draws. Orbit rejection and graph-density mixtures also restrict the sampled distributions. Counts from different systems cannot be pooled into a cross-domain effect.

Measurement and mechanism. CKA, neighbor overlap, and Procrustes emphasize different geometry and have different invariances. Gold-answer controls are informative but are not an input-deconfounding or causal-identification theorem. The study does not establish transfer gains, infer unobserved composite relations, or identify a group-specific internal computation. Matrix row/column identities also hold for general binary matrices and resemble graph-degree relationships. Transpose is an anti-automorphism, while multiplication by a fixed invertible matrix is an action; neither result demonstrates learned group multiplication. Original output-neighbor numerical failures remain a reported reproducibility limitation, not repaired by the paper's wording.

7 Conclusion

Exact mathematical task correspondences are often associated with representation pairing selectivity in the measured settings. What is selected, and what changed through training, depend on the comparison, initial reference, encoding, layer, and control. A raw fixed-transform candidate comparison provides a particularly clear training-origin reference under shared initial weights; answer stratification changes that reference. The full task grid and the matrix F_1/F_2 transpose example with perfect measured task accuracy show why one positive score is insufficient to identify a shared mathematical mechanism. The practical contribution is to make these distinctions explicit and reproducible while retaining their empirical boundaries.

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A Exact task relationships

For a permutation of $[n]$, let $C(\pi)(i) = n + 1 - \pi(i)$ and $I(\pi) = \pi^{-1}$. Table 5 records the full declared catalogue, in its original order. A component is a direct-sum component; a global descent separates an initial segment whose values all exceed the remaining values. The offset in P8 is part of the exact formula, not a change of numeric task labels. The wrong action is complement for the two inverse relations and inverse for the six complement relations. Identity pairing is also retained for every relation.

ID	Left task f	Right task g_*	Exact identity
P1	Descents	Recoils (inverse descents)	$f(\pi) = g_*(I\pi)$
P2	Peaks	Valleys	$f(\pi) = g_*(C\pi)$
P3	Fixed points	Anti-fixed points	$f(\pi) = g_*(C\pi)$
P4	Exceedances	Deficiencies	$f(\pi) = g_*(I\pi)$
P5	LIS length	LDS length	$f(\pi) = g_*(C\pi)$
P6	Longest increasing run	Longest decreasing run	$f(\pi) = g_*(C\pi)$
P7	Left-to-right maxima	Left-to-right minima	$f(\pi) = g_*(C\pi)$
P8	Global descents	Components	$f(\pi) = g_*(C\pi) - 1$

Table 5: All eight directed PermWorld relationships. Tasks are defined by the frozen exact registry; the relationship ID is used in the full tables and figures. Complement-related instances share a transformation family.

GraphWorld uses loop-free directed adjacency matrices A. Reversal is $R(A) = A^\top$ and complement is $C(A) = J - I - A$. Thus maximum out-degree becomes maximum in-degree under R, likewise for minimum

degree; maximum out- or in-degree becomes $n-1$ minus the respective minimum under C. These are four directed instances from two transformation types. Define the fixed wrong comparator W by shuffling the lexicographically ordered unordered vertex pairs with the recorded seed $20261010+n$, swapping successive pairs, and preserving the orientation of their two directed edges. If one pair remains, it stays fixed. The wrong actions are $W \circ R$ and $W \circ C$. The source release includes their exact mapping and per-anchor formula-match Booleans. Graphs are drawn from an off-diagonal Bernoulli mixture with $p \in \{0.25, 0.5, 0.75\}$, subject to disjoint $\langle R, C, W \rangle$ split orbits. A fixed position-wise binary one-hot input kernel is isometric under these slot/category operations and cannot distinguish correct and wrong pairings. This is one static input reference, not a claim that every input feature map has identical geometry.

In MatrixGroupWorld, all matrix arithmetic is over $\mathbb{F}_2$, while wt counts nonzero bits as an ordinary integer. Define

$$F_1(A) = \mathrm{wt}(Ae_1), \quad F_2(A) = \mathrm{wt}(A^\top e_1), \quad F_3(A) = \mathrm{wt}(A(e_1+e_2)), \quad F_4(A) = \mathrm{wt}(A^\top(e_1+e_2)).$$

With $Q = I + E_{21}$ and P swapping coordinates 1 and 2, the original catalogue is

Relation	Directed tasks	Correct $T(A)$	Wrong $U(A)$
R1 (primary)	$F_1 \to F_2$	$A^\top$	$PA^\top$
R2 (supplementary)	$F_3 \to F_4$	$A^\top$	$Q^\top A^\top$
R3 (primary)	$F_1 \to F_3$	AQ	$AQ^\top$
R4 (supplementary)	$F_2 \to F_4$	$Q^\top A$	QA

Each correct action satisfies $g_*(T(A)) = f(A)$. These identities require neither uniform sampling over $\mathrm{GL}(n, \mathbb{F}_2)$ nor a learned group representation. The embedded left/right $\mathrm{GL}(2, \mathbb{F}_2)$ actions, with transpose, yield at most 72 orbit members for split exclusion.

B Complete primary PermWorld decomposition

Table 6 gives all 32 primary conditions. The companion CSV preserves full floating-point precision. There is no omitted negative condition and no inference using the 32 rows as independent repetitions.

C Training and additional measurement details

The declared optimization uses AdamW with learning rate 3×10^{-4}, weight decay 0.01, $\beta = (0.9, 0.999)$, $\epsilon = 10^{-8}$, 1,000 warmup updates, minimum learning-rate ratio 0.1, and gradient norm clipping at 1.0. Microbatches of sixteen accumulated four times give an effective batch of sixty-four. Parameters use FP32 and CUDA BF16 autocast when available; TF32 is disabled. The vocabulary has 188 tokens and maximum sequence length is 128. The causal answer-plus-EOS objective supervises two tokens per example in either permutation encoding.

Balanced random size-permutation blocks determine the saved update schedule. All specialists within a world use the same sequence of anchor IDs. Original four-task PermWorld runs retain their recorded task-specific starting random states; the twelve additional tasks reuse the saved architecture-specific descents starting state. Encoding pairs share the exact applicable stream. Thus parameter sharing is exact while a claim that all sixteen tasks had distinct independent random streams would be incorrect. Existing source versions and engineering corrections are individually recorded in the checkpoint receipts.

Each score is evaluated at the logical input end before the task query; this may be a different absolute token position under the two encodings. The embedding measurement at this position is not a pooled representation of all input tokens. All declared intermediate layers are retained for their prescribed CKA analyses. Static positional/input and gold-answer references have no fictitious trained-minus-initial increment.

In each comparison, answer means are computed separately on fitting and held-out strata, according to the original oracle measurement protocol. Procrustes fitting uses validation data, rank $\min(8, d_f, d_g)$ when

		S_c^0	S_u^0	S_c^t	S_u^t	B_0	B_t	δ_c	δ_u	Δ
P1	TF / line	+0.066	+0.671	+0.119	+0.026	−0.605	+0.093	+0.053	−0.645	+0.698
P1	TF / cycle	+0.812	+0.027	+0.047	+0.079	+0.785	−0.032	−0.765	+0.052	−0.817
P1	GRU / line	+0.079	+0.696	+0.103	+0.036	−0.617	+0.068	+0.024	−0.660	+0.684
P1	GRU / cycle	+0.817	+0.016	+0.109	+0.030	+0.801	+0.079	−0.708	+0.014	−0.722
P2	TF / line	+0.672	+0.072	+0.350	+0.012	+0.600	+0.338	−0.322	−0.060	−0.262
P2	TF / cycle	+0.028	+0.808	+0.067	+0.010	−0.780	+0.057	+0.039	−0.798	+0.837
P2	GRU / line	+0.698	+0.081	+0.886	+0.012	+0.617	+0.874	+0.188	−0.069	+0.257
P2	GRU / cycle	+0.017	+0.816	+0.527	+0.012	−0.799	+0.514	+0.510	−0.803	+1.313
P3	TF / line	+0.670	+0.059	+0.091	+0.011	+0.611	+0.080	−0.580	−0.048	−0.531
P3	TF / cycle	+0.040	+0.661	+0.013	+0.014	−0.622	−0.002	−0.027	−0.647	+0.620
P3	GRU / line	+0.697	+0.072	+0.107	+0.024	+0.624	+0.083	−0.590	−0.048	−0.542
P3	GRU / cycle	+0.028	+0.516	+0.014	+0.026	−0.487	−0.012	−0.014	−0.490	+0.475
P4	TF / line	+0.067	+0.676	+0.129	+0.037	−0.610	+0.092	+0.062	−0.639	+0.701
P4	TF / cycle	+0.650	+0.039	+0.128	+0.011	+0.612	+0.117	−0.522	−0.028	−0.494
P4	GRU / line	+0.080	+0.703	+0.177	+0.132	−0.623	+0.045	+0.097	−0.571	+0.669
P4	GRU / cycle	+0.499	+0.028	+0.131	+0.021	+0.471	+0.110	−0.369	−0.007	−0.361
P5	TF / line	+0.671	+0.049	+0.216	+0.036	+0.622	+0.180	−0.455	−0.013	−0.442
P5	TF / cycle	+0.027	+0.807	+0.130	+0.081	−0.780	+0.049	+0.104	−0.726	+0.829
P5	GRU / line	+0.698	+0.067	+0.517	+0.062	+0.631	+0.455	−0.181	−0.005	−0.176
P5	GRU / cycle	+0.016	+0.811	+0.404	+0.161	−0.795	+0.242	+0.388	−0.649	+1.037
P6	TF / line	+0.670	+0.068	+0.394	+0.018	+0.602	+0.376	−0.276	−0.050	−0.226
P6	TF / cycle	+0.027	+0.812	+0.137	+0.010	−0.784	+0.127	+0.110	−0.801	+0.911
P6	GRU / line	+0.696	+0.078	+0.788	+0.013	+0.619	+0.775	+0.092	−0.064	+0.156
P6	GRU / cycle	+0.016	+0.816	+0.033	+0.136	−0.800	−0.104	+0.016	−0.680	+0.696
P7	TF / line	+0.686	+0.051	+0.211	+0.127	+0.634	+0.084	−0.475	+0.076	−0.551
P7	TF / cycle	+0.024	+0.797	+0.110	+0.053	−0.773	+0.056	+0.085	−0.744	+0.829
P7	GRU / line	+0.698	+0.079	+0.487	+0.073	+0.619	+0.414	−0.212	−0.006	−0.205
P7	GRU / cycle	+0.016	+0.815	+0.236	+0.086	−0.799	+0.150	+0.221	−0.729	+0.950
P8	TF / line	+0.674	+0.059	+0.073	+0.019	+0.614	+0.054	−0.601	−0.040	−0.560
P8	TF / cycle	+0.025	+0.790	+0.028	+0.040	−0.765	−0.012	+0.003	−0.750	+0.753
P8	GRU / line	+0.695	+0.075	+0.489	+0.059	+0.620	+0.430	−0.206	−0.016	−0.190
P8	GRU / cycle	+0.015	+0.808	+0.032	+0.163	−0.793	−0.131	+0.017	−0.645	+0.662

Table 6: Joint-answer-controlled, final-normalized ordinary CKA, computed per length before equal weighting. Values are rounded to three decimals only for presentation; formulas and exported tables retain full precision. "Cycle" means canonical cycle. The original identity pairing and other controls remain in the public source tables.

supported, PCA bases and RMS scales estimated on the fit split, then an orthogonal alignment. A rank-deficient fit is reported undefined rather than reducing the declared rank to obtain a value. The score is the negative held-out normalized RMS error, not CKA on aligned features and not a task-transfer accuracy.

PermWorld and GraphWorld output references use conditional probabilities over numeric answers 0–30; MatrixGroupWorld retains its original full 188-token output softmax. These distinct baseline constructions are not pooled as a common cross-system effect. Source answer accuracy is reported over the full vocabulary. Length/size mode baselines use training labels with the original smallest-label tie rule, rather than fitting a test-set mode. The public tables include the per-size values and source learning scores, including weak specialists and undefined baseline measurements.

D Numerical verification and source availability

MatrixGroupWorld retains its original `completed_with_verification_caveat` status: eight actual 20k endpoints, numerical status counts 64 passed, 2 failed, and 6 `undefined_consistent`, with `scientific_analysis_complete=false`. Both failures concern the trained raw *output-probability* kNN score in unit 1, R1, $n = 6$, on the fixed 64-input subset. The production and independent values differ by 1/640: 0.600000 versus 0.6015625 for the correct view and 0.1656250 versus 0.1671875 for the wrong view. Both exceed the original 10^{-8} tolerance. A read-only diagnosis localizes near-onehot distance roundoff without changing the metric, values, tolerance, or failed status. Those two failed records concern output probabilities. The hidden-CKA checks on the fixed subset record no failures; the audit does not independently verify all 432 production cells.

Code, complete tables, protocols, checkpoint/source identities, and actual data and array packages are available at https://github.com/XuanyuYang223/2026TMLR and its representation-selectivity artifact release.

The paper reads saved tables at publication/saved-table snapshot `d9a113d`; actual execution versions remain recorded separately in the source receipts. Its figures are reproducible with the companion CSV-only plotting script. The publisher separately verified all 89 local archive streams and member SHAs and all public provider sizes and digests. A fresh public clone actually downloaded thirteen artifact packages and independently checked the main saved-table aggregates. These are engineering and arithmetic checks, not new statistical replications or full model-forward reproductions. Historical constructor-rebuilt or unverified initials are kept separate from the genuine initials used in the primary study. Seven historical pre-repair source revisions remain unavailable; no exact replay of every historical development state is claimed.

The additional workshop bibliography item remains a draft task pending verified title/author/venue information. Before a formal anonymous submission, author-identifying repository links should be handled according to the chosen submission policy. This document is publicly shared as a working draft and has not been formally submitted for publication.